# Proportions and percentage change

Lecture and worked examples

# Proportion and proportion of a proportion

In a nonempty finite set $E$, the proportion of a subset $F$ is $p = \mathrm{Card}(F)/\mathrm{Card}(E)$; $0 \leq p \leq 1$. Its expression as a percentage is $100p\,\%$. If $G \subseteq F \subseteq E$, $F \neq \varnothing$, and if $p_1$ is the proportion of $F$ in $E$ and $p_2$ that of $G$ in $F$, the proportion of $G$ in $E$ is $p_1 p_2$.

# Proportion and proportion of a proportion

A school has 1500 students, of whom 652 are day boarders. Calculate their proportion. In a class, 60% of the students play sport; among them, 40% are boys. What proportion of the class do the boys who play sport represent?

# Worked answer: Proportion and proportion of a proportion

$652/1500 \approx 0.4347$, that is about 43.47%. For the class, $0.60 \times 0.40 = 0.24$, that is 24%. The second percentage has the sporty students as its reference, not the whole class.

# Proportion and proportion of a proportion

In a fictional school of 800 students, 35% play a sport at a club; among them, 45% play a team sport. Calculate the two counts and the final proportion.

# Worked answer: Proportion and proportion of a proportion

$800 \times 0.35 = 280$ students at a club. $280 \times 0.45 = 126$ in a team sport. The final proportion is $126/800 = 0.1575 = 15.75\%$.

# Proportion and proportion of a proportion

Proportion and rate of change do not answer the same question: a proportion compares a part to a whole, a rate compares a change to an initial value.

# Absolute change and rate of change

For an initial value $V_I > 0$ and a final value $V_F > 0$, the absolute change is $V_F - V_I$; the rate of change is

$$t = \frac{V_F - V_I}{V_I}.$$

The rate has no unit. It is positive for an increase, negative for a decrease. The reference is always the initial value.

# Absolute change and rate of change

A cheese is sold for 2.13 euros instead of 3.55 euros. Calculate the absolute change and the rate.

# Worked answer: Absolute change and rate of change

Absolute change: $2.13 - 3.55 = -1.42$ euro. Rate: $-1.42/3.55 = -0.40$, that is a decrease of 40%.

# Absolute change and rate of change

$CM = V_F/V_I = 1 + t$, so $V_F = V_I(1 + t)$. To recover $V_I$, divide $V_F$ by $1 + t$ (nonzero).

# Absolute change and rate of change

A fictional item priced 50000 Ar increases by 12%, then separately consider a 12% discount on the initial price. Give the final prices. A price after a 20% increase is 72000 Ar; find its initial price.

# Worked answer: Absolute change and rate of change

Increase: $50000 \times 1.12 = 56000$ Ar. Discount: $50000 \times 0.88 = 44000$ Ar. Last question: $V_I = 72000/1.20 = 60000$ Ar.

# Successive changes

Successive changes with rates $t_1, t_2$ have overall multiplier

$$CM_G = (1 + t_1)(1 + t_2), \qquad t_G = (1 + t_1)(1 + t_2) - 1.$$

We multiply the multipliers; we do not simply add the rates.

# Successive changes

A manufacturer sold 5600 bicycles in 2022, which is 25% more than in 2021. Its sales decrease by 15% in 2023. Determine the sales of 2021 and 2023, then the overall rate between these two years.

# Worked answer: Successive changes

2021: $5600/1.25 = 4480$ bicycles. 2023: $5600 \times 0.85 = 4760$ bicycles. Overall multiplier $1.25 \times 0.85 = 1.0625$, so an overall increase of 6.25%. Check: $(4760 - 4480)/4480 = 0.0625$.

# Successive changes

A price increases by 10% then decreases by 10%. Does it return to the initial price?

# Worked answer: Successive changes

$CM_G = 1.10 \times 0.90 = 0.99$. The final price is 99% of the initial price: an overall decrease of 1%. The two rates refer to different reference values.

# Successive changes

A rate goes from 20% to 25%. Explain the difference between the increase in percentage points and the relative change.

# Worked answer: Successive changes

The gap is 5 percentage points. The relative change is $(0.25 - 0.20)/0.20 = 0.25$, that is an increase of 25%.

# Reverse change

To return to the initial value after a change of rate $t > -1$, use the multiplier $1/(1+t)$. The reverse rate is

$$t_{\text{reverse}} = \frac{1}{1+t} - 1 = \frac{-t}{1+t}.$$

# Reverse change

Which rate compensates a 25% increase? Which rate compensates a 20% decrease?

# Worked answer: Reverse change

After a 25% increase, the reverse multiplier is $1/1.25 = 0.8$: a 20% decrease. After a 20% decrease, it is $1/0.8 = 1.25$: a 25% increase.

# Reverse change

A price undergoes a 30% discount, then a 10% increase. What additional percentage increase restores the initial price?

# Worked answer: Reverse change

The current multiplier is $0.70 \times 1.10 = 0.77$. The required multiplier is $1/0.77 = 100/77$. The rate is $23/77 \approx 0.2987$, that is about 29.87%. Check $0.77 \times (100/77) = 1$.